

SMART INDIA HACKATHON 2026 · PROBLEM STATEMENT SIH26102

SARVSAKSHI

*Final Master Plan — best of **TRILOK idea.pdf**, **MPLADS Nexus roadmap**, and our SARVSAKSHI design.
Evidence-driven integrity layer for proposed, ongoing and completed MPLADS works.*

AI recommends. Authorized officers decide. Investigation priority — never a legal finding of fraud.

Item	Decision
Product name	SARVSAKSHI (■■■■■■■■■■) — every witness agrees
English subtitle	MPLADS Project Integrity & Investigation Layer
Problem	SIH26102 — anomalies, fraud signals, inefficiencies in MPLADS
Organisation	MoSPI / eSAKSHI ecosystem (layer on top, not a replacement)
Sources merged	idea.pdf (TRILOK 6 modules) + MPLADS Nexus (29-page blueprint) + our plan
Status	FROZEN product plan for team execution

1. What we compared

Three inputs were merged. None was copied wholesale. Each had strengths; each had gaps. This document freezes the winning combination.

Source	Strengths we keep	Weaknesses we drop / fix
idea.pdf (TRILOK)	Clear 6-module build list; Need-Gap for MP; Hybrid Risk (rules + peers + explainability); Vision+map footprint; 500 m citizen live photo; milestone gate UI; RAG dossier; 4 demo cases; practical weekly coding order	Wording that 'blocks PFMS disbursement'; SHAP as mandatory for tiny MVP; thin governance; risk of sounding like a payment controller
MPLADS Nexus	Full lifecycle; investigation priority + evidence confidence; Plan-vs-Claim-vs-Evidence; relationship graph; Digital Passport; Need & Impact Index (not 'poor' labels); Copilot guardrails; P0/P1/P2 scope; validation; risks; demo story; human-in-loop on money	Very large; easy to overbuild; relationship graph can steal time from demo if not scoped
Our SARVSAKSHI plan	Three-world check; amber over-billing (size vs bill); photo details (L×W); cross-scheme uniqueness; challenge photo; Jan-Sakshi options+ratings; SC/ST equity helper; DISHA brief; soft sanction/payment language	Can sprawl into too many named modules if not prioritised

Table: Source comparison — keep / drop.

2. Frozen one-line pitch

SARVSAKSHI evaluates MPLADS works from proposal to completion using regional peers, rules, plan–claim–evidence checks, geospatial and citizen signals, then gives officers an explainable investigation priority and next verification step — humans decide.

3. Positioning (from Nexus + our rule)

- Not another MPLADS transparency dashboard (eSAKSHI already monitors).
- Not SIH26103 project-monitoring clone.
- Not a fraud conviction engine and not an autonomous sanction/payment system.
- An **intelligence + verification layer** around eSAKSHI / PFMS workflows.
- Same Project Digital Passport for future, ongoing and completed works; risk evolves as evidence arrives.

4. Final product modules (merged 8)

idea.pdf had 6 modules. Nexus had more screens. We freeze **8 modules**: the TRILOK six, plus Nexus's Passport/Graph spine, plus our citizen options+ratings and soft milestone advisor.

#	Module	Best taken from	What it does
1	Need & Impact Priority	idea Need-Gap + Nexus Need Index + our pre-sanction	Rank proposed works by infrastructure gap, reach, history, feasibility. Suggest only — MP/DA recommend & sanction.
2	Hybrid Risk Pipeline	idea Hybrid Risk + Nexus engines + our region/type peers	Cost, time, overlap, rules → fused investigation priority + evidence confidence + why / why-not-flagged.
3	Plan–Claim–Evidence + Vision	Nexus Plan/Claim/Evidence + idea Vision/OCR + our size/details	Blueprint/BOQ/dimensions vs claims vs photo (reuse/AI/morph signals) vs map footprint/satellite consistency.
4	Relationship Graph	Nexus signature feature	Cluster works via agency, place, type, amount, timing, similarity — catch patterns one row misses.

#	Module	Best taken from	What it does
5	Geofenced Jan-Sakshi	idea 500 m live photo + Nexus citizen + our options/ratings	GPS $\leq$ 500 m, live camera, watermark + lat/long, structured options + 1–5 ratings + optional text.
6	Milestone Clearance Advisor	idea gate UI + Nexus Proceed/Hold/Inspect	Before next tranche: recommend Proceed / Hold / Inspect. Officer decides. No autonomous PFMS control.
7	Investigation Copilot	idea RAG dossier + Nexus Copilot guardrails	Evidence-grounded Q&A; only on stored facts and rulebook snippets. No invented fraud.
8	Digital Passport + DISHA Brief	Nexus Passport + our DISHA pack	Lifecycle record + printable inspection/meeting pack for the Collector.

Table: Final eight modules.

5. Lifecycle workflows

FUTURE / PROPOSED Need & Impact rank → peer cost/time → compliance → overlap/graph → Pre-sanction investigation priority → Officer / MP review (suggest only) ONGOING Milestone claim + docs/blueprint + progress photos → authenticity + plan/claim/evidence + geo + Jan-Sakshi → Risk + Evidence Confidence update → Recommend Proceed / Hold / Inspect → Officer decides → next milestone COMPLETED Full anomaly + final plan/claim/evidence + geo + citizen aggregate → Integrity assessment → Passport + audit trail + DISHA / inspection brief
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Figure: One passport, three stages (from Nexus; payment language softened from idea.pdf).

6. Analytics engines (final)

Do not train a supervised ‘fraud classifier’. There are no reliable national fraud labels. Use peers + unsupervised residuals + rules + evidence objects.

Engine	Method (final)	Output
Cost	Peer group by region + work type (+ scale/rural-urban if available); robust stats; optional Isolation Forest on residuals; optional SoR/material index with date/source	Peer deviation + comparable list
Time / schedule	Peer duration distributions + simple expected-completion forecast	Delay / slippage signal
Overlap / duplicate	Sentence embeddings + place/amount/date features; GPS near-duplicates; optional cross-scheme near-match	Linked works + similarity reasons
Compliance	Transparent rule engine from MPLADS Guidelines 2023 (45-day SLA, spend vs sanction, SC/ST helpers, prohibited patterns)	Rule IDs + guideline refs — NOT an ML model
Relationship graph	NetworkX MVP: agency / place / type / timing / amount / similarity edges	Cluster risk
Evidence intelligence	Perceptual hash / visual similarity; EXIF/GPS; optional AI-image weak signal; OCR on BOQ/blueprint; footprint vs billed area when resolution allows	Authenticity / consistency signals
Risk fusion	Weighted or calibrated ensemble over evidence objects; separate Evidence Confidence score	0–100 investigation priority + confidence + SHAP or rule-trace explanation
Citizen aggregate	Options counts + trimmed-mean ratings; geo/time/session trust; rate limits	Supporting Jan-Sakshi signal

6.1 Example fusion weights (prototype only — not MoSPI official)

Component	Weight	Note
Cost	25%	Regional peer deviation
Schedule	15%	Delay / slippage
Overlap	15%	Semantic + spatial
Compliance	15%	Triggered rules
Evidence (plan/photo/geo)	15%	Inconsistency / quality
Relationship graph	10%	Cluster patterns
Citizen	5%	Trusted aggregates only

UX must show: **Investigation Priority** and **Evidence Confidence** separately (from Nexus). Example: Priority 88 + Confidence 46 → “High attention, evidence incomplete — request inspection,” not “Fraud 88%.”

7. Need & Impact Priority (not ‘poor places’)

idea.pdf’s planning engine is kept. Nexus’s language is adopted: do **not** label places only as developed / developing / poor. Build a transparent Need & Impact Index.

- Signals: infrastructure gap, population/reach, access, prior similar investment, sector policy weight, urgency, feasibility (cost/time/compliance readiness).
- Output: ranking aid for remaining entitlement — **not autonomous sanctioning**.
- Optional equity helper: SC/ST 15% / 7.5% placement check (internal), separate from the need index.

8. Plan vs Claim vs Evidence (standout verification)

Layer	Inputs	Question
Plan	Sanctioned scope, blueprint, dimensions, estimate, milestones	What should exist?
Claim	Reported progress, amount used, completion statement	What is being claimed?
Evidence	Docs, photos+metadata, geo/satellite, Jan-Sakshi, inspection	What can be checked?

8.1 Amber over-billing case (from our plan — keep)

Real photo of a smaller/cheaper building with a higher bill is not a ghost. Photo details (L×W, floors, roof) + peer cost + footprint catch **cost–size mismatch**. Colour: amber.

8.2 Satellite honesty rule (from Nexus)

Use imagery for location and rough footprint when resolution allows. If not defensible, return **verification inconclusive** — never invent exact architectural measurements.

9. Jan-Sakshi (citizen) — final rules

- Must be within **500 m** of the work (idea.pdf).
- Live capture preferred (no gallery) when feasible; watermark + lat/long + timestamp.
- Structured **options** (empty / smaller / as claimed / locked / wrong place / ongoing / other).
- Ratings 1–5: existence, matches claim, public use, official photo matches place.
- Optional textarea for mistakes / notes; NLP can categorise themes (idea/Nexus).
- Supporting signal only — never automatic truth; aggregate independent reports (Nexus).
- No public MP star leaderboard; hide personal identity from officers where possible.

10. Milestone Clearance Advisor (final wording)

idea.pdf’s Clearance Gate UI is excellent for demos. Nexus’s governance is mandatory for MoSPI: the system **recommends** Proceed / Hold / Inspect; the authorized officer records the decision. Prototype does **not** autonomously release government funds.



11. Investigation Copilot (not a generic chatbot)

- Tied to the open project’s evidence store and guideline snippets only.
- Answers: why high risk, peers, triggered rules, missing evidence, inspector checklist, why-not-flagged.
- Guardrails: no invented documents/dates/legal guilt; say inconclusive when evidence is thin.
- Log model/prompt/evidence-set versions for important answers.

12. Core screens

Screen	Primary user	Job
Officer Dashboard	DA / MoSPI	Priorities, colours, search, trends
Need & Impact Board	MP / DA	Pre-sanction ranking aid
Project Digital Passport	Officer	Full lifecycle: finance, plan, risk timeline, evidence, decisions
Investigation Workspace	Inspector	Peers, graph, evidence, confidence, actions
Evidence Portal	IA / inspector	Upload docs, blueprint, progress photos, claims
Jan-Sakshi Portal	Citizen	Geo-fenced options + ratings + live photo
Milestone Advisor	Officer	Proceed / Hold / Inspect recommendation
Copilot drawer	Officer	Evidence-grounded Q&A;
DISHA / Brief PDF	Collector	Meeting pack / inspection note

13. Finale demo story (merged)

One connected story (Nexus) using four planted cases (idea.pdf colours).

Step	Show
1 Future	Propose a work → Need & Impact + pre-sanction peer risk
2 Explain	Comparable regional works + why unusual / why-not-flagged
3 Graph	Linked works (same agency / place / timing)
4 Passport	Investigation Priority + Evidence Confidence
5 Evidence	Blueprint + progress photo → Plan vs Claim vs Evidence
6 Geo	Location / footprint check with honest confidence
7 Citizen	Aggregated 500 m Jan-Sakshi (options + ratings)
8 Copilot	Why flagged? What should inspector verify?

Step	Show
9 Decision	Hold / Inspect / Proceed + reason → audit trail
10 Close	Same passport continues through milestones / completion

13.1 Four planted demo works (must memorise)

Case	Colour	Story
A Ghost	Red	Marked complete; reused/wrong photo; empty ground; locals say empty
B Over-bill	Amber	Real photo; small footprint / tin vs RCC claim; peer cost high
C Stuck	Blue	45-day / long delay with bottleneck cause — inefficiency
D Clean	Green	Peers OK; plan/claim/evidence agree; locals satisfied

14. Build scope: P0 / P1 / P2

Priority	Must include	Role
P0 — Demo critical	Real/public MPLADS-shaped data; search; cost/time anomaly; overlap; rules; risk fusion; evidence objects; Passport; peers; Plan-vs-Claim-vs-Evidence (sample docs); human review; 4 demo cases	Cannot cut
P0 — Standout	Evidence Confidence; relationship graph (MVP); Investigation Copilot (grounded); Need & Impact board	Differentiation
P1 — Strong if time	Image duplicate/AI weak signals; geo consistency; milestone advisor UI; Jan-Sakshi 500 m portal	Pilot depth
P2 — Roadmap only	Full live satellite CV measurements; live PFMS control; production security; continuous retraining; deep cross-scheme APIs	Say 'next' on PPT

Team rule (from Nexus): a feature belongs in the core demo only if it visibly improves an explainable decision.

15. Recommended tech stack (merged)

Layer	Choice	Why
Frontend	React or Next.js + Tailwind (navy/saffron) + charts + Leaflet	idea.pdf UI speed
Backend	Python FastAPI	ML + rules orchestration
DB	SQLite (MVP) → PostgreSQL later	Hackathon shareability
ML	pandas + scikit-learn (± Isolation Forest); optional LightGBM	Nexus honesty
NLP	Sentence Transformers + cosine	Overlap
Graph	NetworkX	Relationship intelligence
Docs/OCR	pdfplumber/PyMuPDF + Tesseract	Blueprint/BOQ fields
Vision	OpenCV + imagehash; optional CLIP/YOLO offline	Roof/type + reuse
Maps	Leaflet + OSM; Bhuvan/satellite where licensed	No key for basic demo
Copilot	LangChain/RAG over evidence + guideline PDF; or template explanations	Grounded only
Deploy	Docker optional; local demo first	Reproducible

16. Data strategy

- Primary: public MPLADS / eSAKSHI-shaped work records with provenance (URL, extract date).
- Guidelines 2023 → rule engine; CAG patterns → validation cases, not fake fraud labels.
- Synthetic anomalies clearly marked for testing.
- Citizen/IA uploads are platform-generated evidence during pilot.
- Never claim live ministry private login data if you do not have it.

17. Hard governance rules (frozen)

- Never say ‘this is fraud’ or print a fraud probability as a legal finding.
- Never autonomously sanction works or release funds in the prototype.
- LLM never invents evidence; never unrestricted raw dump into the model.
- Citizen one report ≠ truth; require trust signals + aggregation.
- Satellite inconclusive when imagery cannot support the claim.
- Version rules, models, prompts; audit officer decisions.
- Label demo/synthetic rows in presentations.

18. Team ownership (practical)

Owner	Owns
Data	Ingest, clean, provenance, peer groups, demo cases
ML	Cost/time/risk fusion, calibration, explanations
NLP/Graph	Similarity, relationship graph
CV/Geo	Docs/OCR, image checks, map/footprint
Backend	FastAPI, evidence API, passport, Copilot wiring
Frontend	Dashboard, passport, citizen, milestone UI, demo polish
All	One story; rehearse 4 cases; PPT; viva answers

19. What not to claim

- 97% fraud accuracy / trained on millions of labelled MPLADS fraud cases.
- We replace eSAKSHI, CAG, or PFMS.
- We auto-sanction MPs’ works or auto-pay vendors.
- Exact satellite dimensions when resolution is insufficient.
- AI-generated-image detector is perfect proof.
- Live ministry data when using samples.

20. Closing line for judges

We do not stop at ‘this project looks unusual.’ We show the peers, the connected pattern, the plan–claim–evidence chain, the confidence of that evidence, the citizen corroboration, and the next verification step — then leave the decision with the authorized human.

This document freezes SARVSAKSHI as the team’s final product definition. Build order and prototype cuts follow Section 14 (P0/P1/P2). Do not reopen ChatGPT-only dashboard ideas or autonomous payment claims.

Source PDFs: Documents\SIH\idea.pdf (TRILOK roadmap) and Downloads\MPLADS_Nexus_Complete_Project_Roadmap.pdf — merged with our prior SARVSAKSHI design.